

Filing Edge

Systematic Trading | Research and decision system

Executive summary

A reproducible research desk that connects corporate disclosures to options, costs, capacity and a documented allocation decision. Its useful output can be a rejection: research results are not converted into a trading signal without evidence.

Development: 13 complete pairs, difference -186.2 bps, interval -427.3 to +74.2. Held-out test: 5 pairs, difference +319.9 bps, interval +2.9 to +815.4. Intervals are exploratory envelopes of issuer and calendar-quarter bootstraps (1,999 draws each), not joint two-way cluster confidence intervals.

Economic foundation

Selling downside protection earns premium in exchange for a contingent purchase obligation. The other side may value insurance or immediate liquidity. A buyback disclosure might change the distribution of future returns or insurance demand, but authorization does not ensure actual repurchases. This is a falsifiable conditional risk-premium hypothesis, not a claim of arbitrage.

Innovation and product value

The system joins exact filing accessions with contract selection, timestamped quotes, event/control comparisons and funded capital accounting. It exposes missingness and preserves the original pilot. Gemini explains exact source excerpts with quote validation; Tiger Data stores experiments and serves precomputed daily activity; ElevenLabs narrates the computed decision with a visible transcript.

Current allocation decision

RESEARCH ONLY. Historical tag availability is unverified; one-session availability is assumed. A static 2026 universe introduces selection bias. The spot proxy uses American put-call parity with daily closes, 4% interest and no dividend adjustment. Early assignment is not reconstructed. Pair deletion for missing quotes can bias results. These limits prevent live allocation.

Data and methodology

Data contract

Massive supplies 8-K disclosures, option reference contracts, option daily trades and historical NBBO. Raw licensed responses and credentials remain local. Only standard contracts are included. Results are truncated at each window end; future labels and exits are not silently filled.

Fixed 100-issuer universe; buyback tag share_repurchase_program. The primary trade sells a standard, 100-share, cash-secured put, 5% OTM at the pre-filing parity reference, in the 90-180 day bucket nearest 120 days. Entry is anchor +1 session; primary exit is anchor +21. Every fixed horizon is reported. Contract selection uses pre-anchor data.

Execution and matching

At 15:55 Eastern (12:55 on early closes), use the latest NBBO no more than 120 seconds old. Require positive sizes/prices and a spread at most 25% of midpoint. Sell at bid less 5%; buy at ask plus 5%; charge \$0.65 per contract per side. Returns divide by full strike collateral. Quotes indicate prices, not guaranteed fills.

Controls use the same issuer at fixed backward offsets of 35, 50, 65 or 80 sessions, excluding nearby buybacks. Choose the closest pre-anchor premium/strike and maturity, with premium ratio at most 1.5 and maturity difference at most 35 days. No future return selects the control; unmatched events remain eligible for portfolio entry.

Time and validation

The filing date maps to the next eligible exchange session; tag availability is assumed one session later. This is not proof of first public dissemination. US exchange holidays and early closes are handled explicitly. Tests exercise quote timestamps, crossed quotes, missing exits, cash conservation, collateral limits and assignment cashflows.

Reproducibility

Use submission/Filing_Edge_Judge.ipynb in a clean Python 3.11+ kernel, or judge.py with --start, --end and --mode sealed. The notebook embeds checksummed source and asks for only a Massive key. Research hashes and first-access manifests are immutable within each workspace.

Results across fixed horizons

Development: 13 complete pairs, difference -186.2 bps, interval -427.3 to +74.2. Held-out test: 5 pairs, difference +319.9 bps, interval +2.9 to +815.4. Intervals are exploratory envelopes of issuer and calendar-quarter bootstraps (1,999 draws each), not joint two-way cluster confidence intervals.

Horizon	Dev pairs	Dev diff bps	Test pairs	Test diff bps
1	0	N/A	0	N/A
2	16	-20.6	6	-138.3
3	17	-13.3	5	-109.7
5	18	-28.3	5	+85.1
10	16	-123.6	6	+271.5
21	13	-186.2	5	+319.9
42	11	-63.1	4	-690.8
63	3	-2.1	3	-348.0
exp	0	N/A	0	N/A

Reading the result correctly

The held-out event return is negative even though its difference from worse-performing controls is positive. Five pairs across only two calendar quarters cannot support a strong allocation conclusion.

The primary horizon is 21 sessions. Other horizons, entry delays of 1/2/3/5 sessions, haircuts of 2.5/5/10%, expiry buckets and OTM levels are exploratory sensitivity checks, not a search for the best result. Full cell counts, intervals and exclusions are exported in JSON.

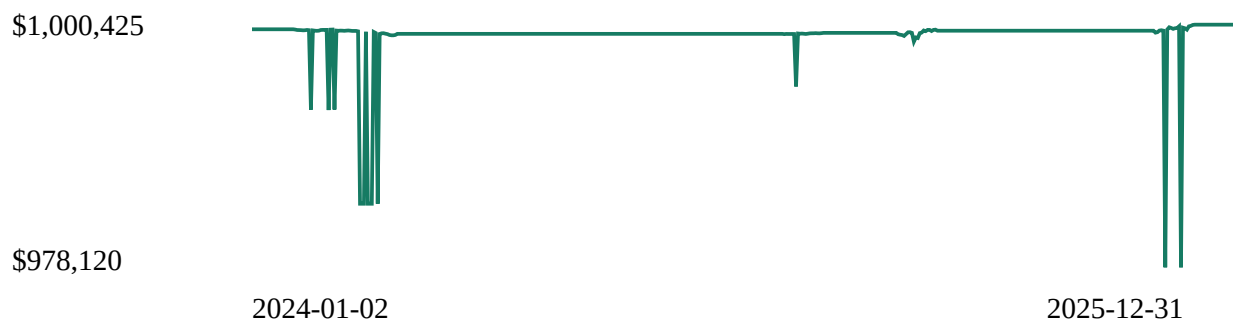
Missingness and confounding

Development covered 36 events; test covered 14. Complete pair counts can vary across cells. Filtering wide or unavailable quotes improves realism but can change the population. Premium and maturity matching does not remove omitted risk factors or establish causality.

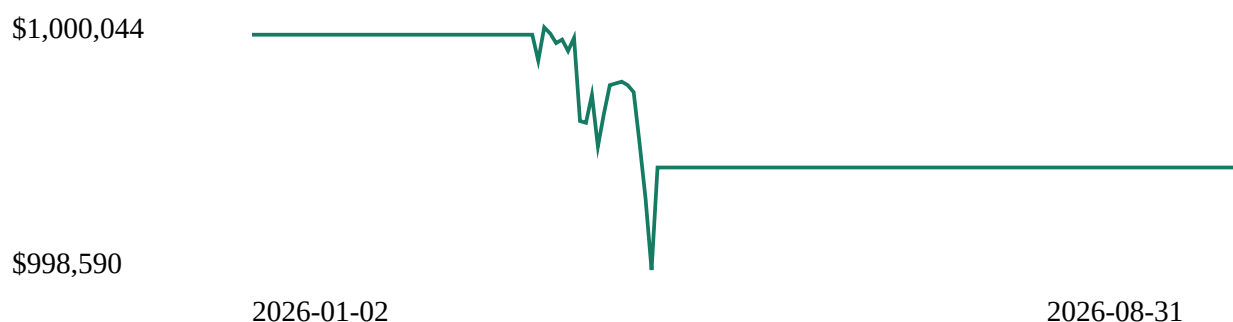
Funded portfolio and risk

Metric	Development	Held-out test
Total account return	0.04%	-0.08%
Annualized return (CAGR)	0.02%	-0.12%
Annualized volatility	4.41%	0.14%
Sharpe, zero cash benchmark	0.03	-0.86
Maximum drawdown	-2.21%	-0.15%
Premium cash turnover / year	0.0025x	0.0047x
Positions admitted	6	1

Development account equity



Held-out account equity



Initial capital is \$1 million in each independent window. Idle cash earns zero. Turnover counts traded option premium, not underlying notional. No leverage. Missing marks use a full-strike liability reserve, a pessimistic accounting bound that can dominate volatility and trigger the entry pause.

Liquidity, limitations and delivery

Capital and capacity

Per-issuer allocation is 5% of initial equity; total reserved strike collateral is at most 50%. Quantity is the minimum allowed by cash, lagged 20-observation option volume (at least five observations), 1% participation and 25% displayed bid size. This is a conservative size constraint, not a calibrated market-impact capacity estimate. Scaling requires depth and fill studies.

Assignment and operational controls

Physical put assignment pays strike times 100 shares per contract and receives shares. Unit-tested scenarios value their liquidation at shocked prices. Actual historical exercise notices, overnight stock execution, dividends, borrow and taxes are absent. Open positions remain reserved at the cutoff. Consequently the model cannot claim a complete historical exercise lifecycle.

What would justify further work

Archive live disclosure arrival times, validate a point-in-time universe, cross-check parity against permitted reference data, reconstruct dividends and assignment, and paper trade quoted orders. Predeclare any revised method before gathering new evaluation data. Do not reuse the evaluated 2026 window as an untouched test.

Submission and sponsor use

The repository contains the application, standalone judge notebook, frozen configuration, tests, deployment files, both research notes and a demo script. API keys stay server-side. Gemini explanations cannot modify signals. ElevenLabs audio is labeled AI-generated and includes its exact report-derived transcript. Tiger Data persists provenance and uses a Timescale continuous aggregate for daily queries.

Sources and audit trail

Challenge: gqhacks.com/tracks/systematic-trading and the supplied GQH Massive 8-K options starter notebook. Data documentation: massive.com/docs. Sponsor documentation: docs.tigerdata.com, ai.google.dev/gemini-api/docs, elevenlabs.io/docs. Source citations in filing evidence link to SEC EDGAR. See `STARTER_AUDIT.md` and `submission/research-manifest.json` for method hashes and access timestamps.